

Writing Logbook — Assignment 2A

CSP529 Technical Writing & Publishing

From L^AT_EX Elements to a Professional Technical Document

MT26MCS024 — Gaurav Acharya MT26MCS032 — Sandeep

September 2026

1. Assignment Overview

Item	Detail
Practical Title	Practical 2A — From L ^A T _E X Elements to a Professional Technical Document
Topic	Restructuring the OS Process document into a proper academic paper
Group Members	Gaurav Acharya (MT26MCS024), Sandeep (MT26MCS032)
Builds Upon	Assignment 02 (Stage 2: Representations of an OS Process)
Date of Completion	September 2026

2. Objectives & Learning Outcomes

This assignment extended Assignment 02 by transforming a collection of isolated L^AT_EX elements into a cohesive, professionally structured technical document. The learning outcomes included:

- Restructuring content into standard academic sections: Title → Abstract → Introduction → Technical Background → Analysis/Working → Discussion → Conclusion → References.
- Mastering floats: captions, labels, and cross-references for figures and tables.
- Making equations part of the argument, not decorative additions.
- Understanding that algorithms describe *procedure* while prose communicates *purpose, rationale, and interpretation*.
- Learning L^AT_EX cross-referencing (`\ref{}`), citation mechanics (`\cite{}`), and bibliography management.

3. Work Performed — Stage by Stage

3.1. Stage 1: Inspecting Previous Work (Practical 2)

We critically examined our Assignment 02 document and identified:

- L^AT_EX constructs used: `figure`, `table`, `equation`, `algorithm`, `itemize`, `enumerate`.
- Elements that were difficult: float placement, equation formulation.
- Elements inserted only because required: some lists felt mechanical rather than communicative.
- Elements that genuinely helped: the equation and algorithm provided clarity that prose alone could not.
- **Critical finding:** Most elements were *not* referred to from the body text — they simply “appeared” without introduction or interpretation.

Key Rule Discovered: *Nothing should appear in a technical document without being introduced, referred to, or interpreted in the text.*

3.2. Stage 2: Building Proper Document Structure

We restructured the entire document into:

1. **Abstract:** A concise overview of the document’s scope — OS boot process, hardware-software interfacing, kernel resource management.
2. **Introduction:** Context and motivation, with an inline citation [?].
3. **Technical Background:** Component comparison table (now with textual reference: “The comparison presented in Table 1 indicates...”) and the state-transition equation (now with interpretation).
4. **Analysis/Working:** Chronological boot sequence (numbered list), algorithmic pseudocode, and the architectural figure (now with: “The computational steps are illustrated in Figure 1”).
5. **Discussion:** Plain-language explanation of the algorithm, plus OS core responsibilities.
6. **Conclusion:** Summary of findings.
7. **References:** Three genuine academic sources.

Justification: Technical Background establishes definitions and models *before* they are used in Analysis. Discussion interprets the formal constructs in accessible language.

3.3. Stage 3: Mastering Figures and Tables

We ensured every float was properly integrated:

- Figure with caption (“Operating System Architecture Layers”), label (`fig:os_process`), and cross-reference from text.
- Table with caption (“System Component and Execution Phase”), label (`tab:os_components`), and cross-reference from text.

Discoveries:

- Figures and tables are numbered automatically by L^AT_EX.
- Inserting a new figure before an existing one automatically rennumbers all subsequent figures.
- `\ref{}` ensures references update automatically; manual numbering would break on any restructuring.
- Captions are essential because they allow figures/tables to be understood independently of surrounding text.

3.4. Stage 4: Mathematics as Part of the Argument

We included two meaningful equations:

1. **Equation 1 — OS State Transition:** $S_{t+1} = k(S_t, \sum I_i + \sum C_j) - \Delta_{overhead}$
Numbered, labelled, cross-referenced, and interpreted: “This equation demonstrates that the OS continuously evaluates the current state alongside hardware interrupts and system calls...”
2. **Equation 2 — Turnaround Time:** $T_{turnaround} = T_{completion} - T_{arrival}$
Added as a second equation to analyse CPU scheduling efficiency, also numbered, labelled, and interpreted.

Answer to Stage 4 question: The equations communicate *quantitative relationships* and *variable dependencies* that the paragraph alone cannot express with the same precision and conciseness.

3.5. Stage 5: Algorithm vs. Explanation

We maintained the OS Boot Sequence pseudocode and added explanatory prose in the Discussion section that translates the algorithmic steps into plain language.

Discovery: Pseudocode describes *what happens step by step* (procedure), while prose communicates *why* each step matters, what assumptions underlie it, and how to interpret the outcomes (purpose and rationale).

3.6. Stage 6: Cross-Referencing Challenge

We added cross-references to: 2 sections, 2 figures/tables, 1 equation, and 1 algorithm. After inserting a new section and figure in the middle of the document and recompiling:

- **What changed automatically:** All section numbers, figure numbers, table numbers, and equation numbers updated throughout the entire document.
- **Why logical references matter:** In a professional document with dozens of figures, manually updating “Figure 3” to “Figure 4” everywhere would be error-prone and unsustainable. `\ref{}` ensures consistency by design.

3.7. Stage 7: References

We included three genuine references connected to our technical content:

1. Silberschatz, Galvin, Gagne — *Operating System Concepts*, 10th ed., 2018.
2. Tanenbaum, Bos — *Modern Operating Systems*, 4th ed., 2014.
3. Love — *Linux Kernel Development*, 3rd ed., 2010.

Skills learned: `\cite{}`, the `bibliography` environment, citation-reference distinction.

3.8. Stage 8: L^AT_EX Debugging Challenge

Error/Problem	What Caused It	How It Was Corrected
Figure not appearing	Missing <code>float</code> package for [H] placement	Added <code>\usepackage{float}</code>
Undefined reference “??”	<code>\label</code> placed before <code>\caption</code>	Moved <code>\label</code> after <code>\caption</code>
Missing package error	Used <code>array</code> column types without importing	Added <code>\usepackage{array}</code>

4. Comparison: Assignment 02 vs. Assignment 2A

Aspect	Assignment 02 (Before)	Assignment 2A (After)
Structure	Flat list of representations	Full academic structure (Abstract to References)
Floats	Inserted without context	Introduced, referenced, and interpreted in text
Equations	Single equation, no interpretation	Two equations, both numbered, labelled, interpreted
References	None	Three genuine academic sources with <code>\cite{}</code>
Cross-references	None	Sections, figures, tables, equations all cross-referenced

5. Key Takeaways

1. The transition from “using L^AT_EX elements” to “writing a technical document” requires every element to serve a communicative purpose within the document’s argument.

2. Cross-referencing is not a convenience — it is a requirement for any document that may be revised or extended.
3. The **Abstract** → **Introduction** → **Background** → **Analysis** → **Discussion** → **Conclusion** structure is not arbitrary; it mirrors the reader’s cognitive needs.
4. Debugging L^AT_EX compilation errors (undefined references, missing packages, float placement) is a core practical skill that improves with experience.

References

- [1] A. Silberschatz, P. B. Galvin, G. Gagne, *Operating System Concepts*, 10th ed., Wiley, 2018.